

0.1 Section 6.2 - Pigeonhole Principle

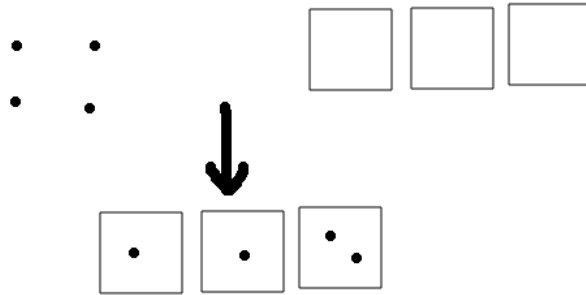
Definition 1 (Pigeonhole Principle)

If there are n boxes and more than n objects placed in these boxes, at least one box contains more than one object.

At least one box will contain

$$\lceil \frac{k}{n} \rceil$$

objects



Example 1

There are 43 people in this room. Are we guaranteed that

1. At least 2 people have the same birth month?

Solution:

Yes, since 12 months ; 43 people